

- Describe **and** explain how a resting potential of an axon membrane is maintained.

[6]

- (b) Hypokalaemia is a condition in which there is a low concentration of potassium ions ( $K^+$ ) in the body. This can affect nervous coordination.

Fig. 7.1 shows a normal action potential and Fig. 7.2 shows an action potential of a person with hypokalaemia.

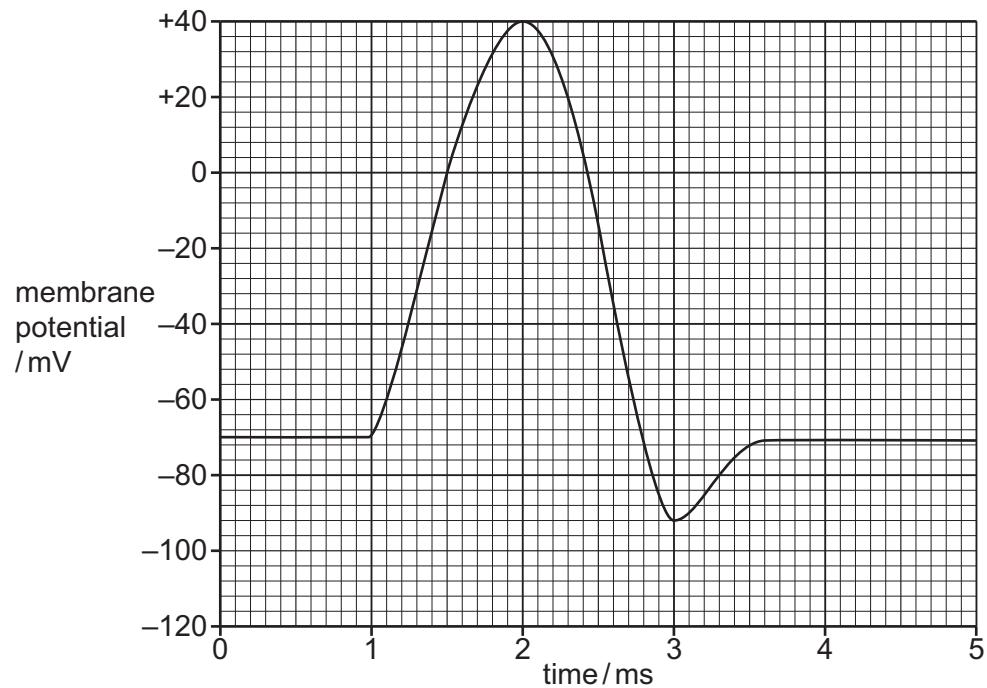


Fig. 7.1

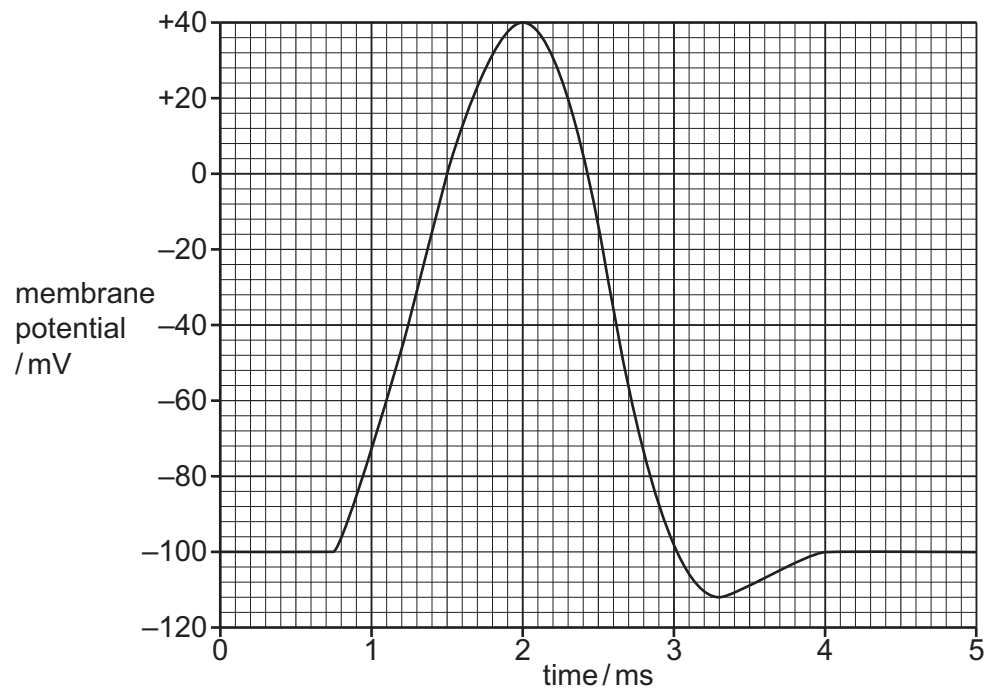


Fig. 7.2



- (i) With reference to Fig. 7.1 and Fig. 7.2, describe the differences between a normal action potential and an action potential of a person with hypokalaemia.

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- (ii) Suggest how hypokalaemia may affect nervous coordination.

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[Total: 11]